St. George's University
Grenada, West Indies

Basic Principles of Medicine I

Module: Cardiovascular, Pulmonary and Renal | Lecture CPR 03

Histology of the Cardiovascular System

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Objectives

SOM.MK.I.BPM1.3.CPR.1.HCB.1113	List the 3 layers of the heart.
SOM.MK.I.BPM1.3.CPR.1.HCB.1114	Describe the structure and function of the 3 layers of the heart (epicardium, myocardium, endocardium).
SOM.MK.I.BPM1.3.CPR.1.HCB.1115	Give the function(s) of Purkinje fibers.
SOM.MK.I.BPM1.3.CPR.1.HCB.1116	Sate the localization of Purkinje fibers.
SOM.MK.I.BPM1.3.CPR.1.HCB.1117	Describe the microscopic structure of Purkinje fibers.
SOM.MK.I.BPM1.3.CPR.1.HCB.1118	Discuss the functions of vascular endothelial cells.
SOM.MK.I.BPM1.3.CPR.1.HCB.1119	Describe the microscopic structure of vascular endothelial cells.
SOM.MK.I.BPM1.3.CPR.1.HCB.1120	List 3 substances released by endothelial cells that prevent thrombus formation.
SOM.MK.I.BPM1.3.CPR.1.HCB.1121	List at least 5 vasoactive substances released by the endothelial cells.
SOM.MK.I.BPM1.3.CPR.1.HCB.1122	Explain how these vasoactive substances influence blood flow.
SOM.MK.I.BPM1.3.CPR.1.HCB.1123	Describe the process of "endothelial activation".
SOM.MK.I.BPM1.3.CPR.1.HCB.1124	Discuss the clinical significance of "endothelial activation".
SOM.MK.I.BPM1.3.CPR.1.HCB.1125	State the functional significance of Weibel Palade bodies
SOM.MK.I.BPM1.3.CPR.1.HCB.1126	Briefly describe the structure of Weibel Palade bodies.
SOM.MK.I.BPM1.3.CPR.1.HCB.1127	Explain the significance of growth factors (VEGF) produced by endothelial cells.
SOM.MK.I.BPM1.3.CPR.1.HCB.1128	Give the function of large (elastic) arteries.
SOM.MK.I.BPM1.3.CPR.1.HCB.1129	Describe the microscopic structure of large (elastic) arteries.
SOM.MK.I.BPM1.3.CPR.1.HCB.1130	Give the function of medium (muscular) arteries.
SOM.MK.I.BPM1.3.CPR.1.HCB.1131	Describe the microscopic structure of medium (muscular) arteries.
SOM.MK.I.BPM1.3.CPR.1.HCB.1132	Give the function of small arteries.

Objectives

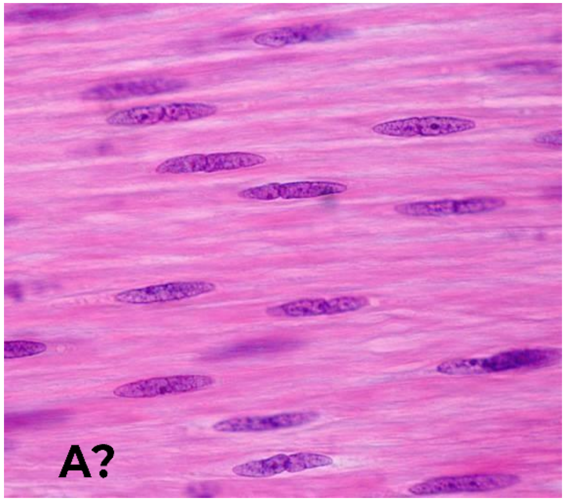
SOM.MK.I.BPM1.3.CPR.1.HCB.1133	Describe the microscopic structure of small arteries.
SOM.MK.I.BPM1.3.CPR.1.HCB.1134	Give the function(s) of arterioles.
SOM.MK.I.BPM1.3.CPR.1.HCB.1135	Describe the microscopic structure of arterioles.
SOM.MK.I.BPM1.3.CPR.1.HCB.1136	Describe the microsopic structure of the three types of capillaries.
SOM.MK.I.BPM1.3.CPR.1.HCB.1137	Identify the three types of capillaries in electron photomicrographs.
SOM.MK.I.BPM1.3.CPR.1.HCB.1138	Give the function of venules (postcapillary and muscular).
SOM.MK.I.BPM1.3.CPR.1.HCB.1139	Describe the microscopic structure of venules (postcapillary and muscular).
SOM.MK.I.BPM1.3.CPR.1.HCB.1140	Give the function of small veins.
SOM.MK.I.BPM1.3.CPR.1.HCB.1141	Describe the microscopic structure of small veins.
SOM.MK.I.BPM1.3.CPR.1.HCB.1142	Give the function of medium sized veins.
SOM.MK.I.BPM1.3.CPR.1.HCB.1143	Describe the microscopic structure of medium sized veins.
SOM.MK.I.BPM1.3.CPR.1.HCB.1144	Give the function of large veins.
SOM.MK.I.BPM1.3.CPR.1.HCB.1145	Describe the microscopic structure of large veins.
SOM.MK.I.BPM1.3.CPR.1.HCB.1146	Describe the microscopic structure of lymphatic capillaries.
SOM.MK.I.BPM1.3.CPR.1.HCB.1147	Describe the microscopic structure of lymphatic vessels.
SOM.MK.I.BPM1.3.CPR.1.HCB.1148	Identify the layers of the vessel wall involved in “arteriosclerosis” and “atherosclerosis”.
SOM.MK.I.BPM1.3.CPR.1.HCB.1149	Define the term “aneurysm”.
SOM.MK.I.BPM1.3.CPR.1.HCB.1150	Identify the layer of the vessel wall affected that lead to the formation of aneurysm.
SOM.MK.I.BPM1.3.CPR.1.HCB.1151	Identify the layers of the vessel wall affected in aortic dissection.
SOM.MK.I.BPM1.3.CPR.1.HCB.1133	Describe the microscopic structure of small arteries.

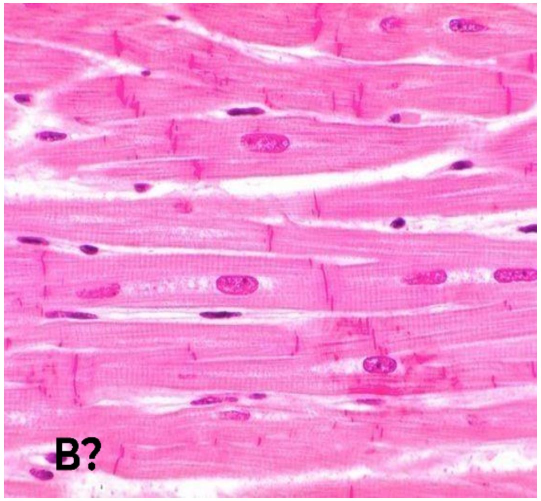
Overview/Summary

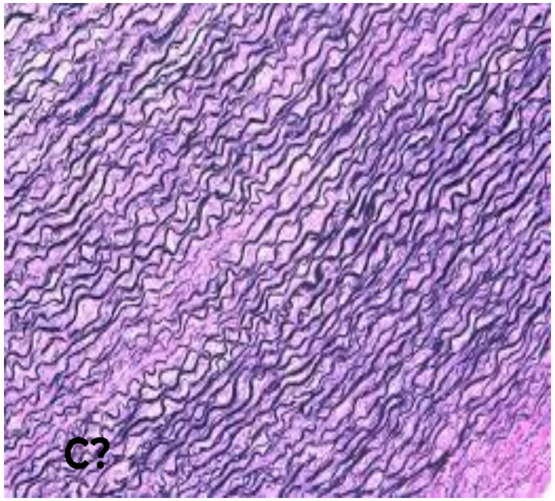
- Components of the Cardiovascular System
  - Heart and its layers
  - Types of blood vessels
    - Features of arteries, capillaries and veins
- Lymphatic channels
- Functions of endothelial cells
- 3 Common diseases of blood vessels

The Cardiovascular System

- Also known as the circulatory system
- Consists of the heart, its related blood vessels and the blood that runs through them
  - The heart is a muscular pump
    - Pumps blood around the body
  - The blood vessels consist of arteries, arterioles, capillaries, venules and veins
    - Carries the pumped blood
  - Blood is a type of specialized connective tissue
    - The cells are called formed elements
    - The extracellular matrix is the plasma
      - Composed of water, proteins, other solutes (eg nutrients)
    - Blood functions to transport (nutrients, gases), regulate (pH buffer) and protect (immunity, hemostasis)
- Basic tissue types seen in this system:







The Heart

Consists of

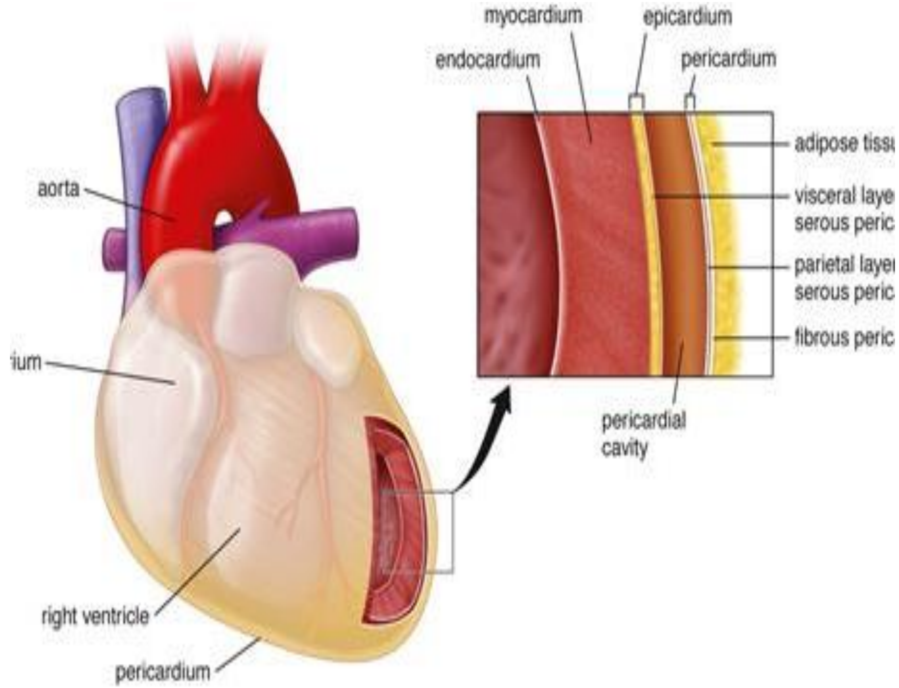
- Muscles – Myocardium
  - For contraction to propel the blood
- A fibrous skeleton
- A conducting system –
  - for initiation and propagation of electrical impulses which results in rhythmic cardiac muscle contractions
- Coronary vasculature –
  - functions in coronary circulation

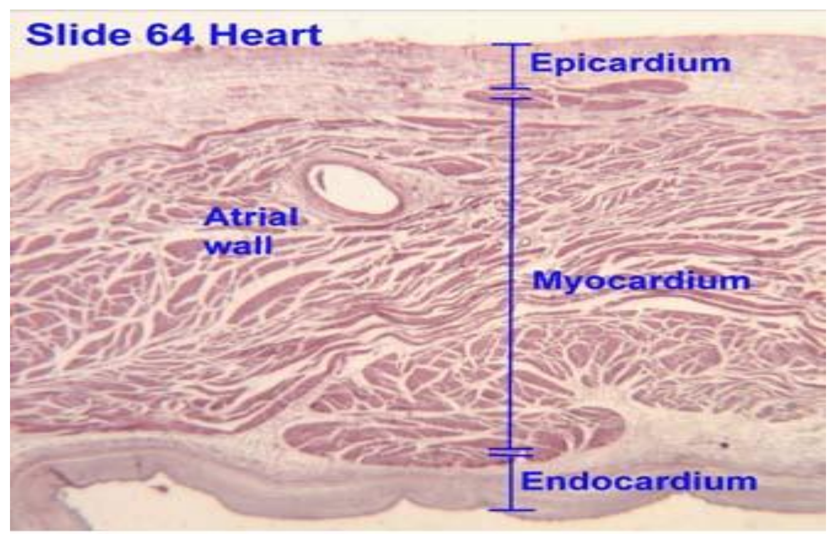
Surrounded by pericardium

- Connective tissue that surrounds and covers the heart walls
- Two portions: fibrous and serous
- Serous layer consists of a parietal layer (which lines the inner aspects of the fibrous pericardium and a visceral layer (which rests on the heart itself)
- Visceral serous pericardium is also called epicardium

Layers of the Heart Wall

- Three layers that share similar characteristics to that found in blood vessels
- **Epicardium (outermost layer)**
  - Also known as visceral pericardium
    - Lined by mesothelium
  - Beneath the visceral pericardium is a layer of adipose tissue
    - Houses the coronary vessels, lymphatics and nerves that supply the myocardium
  - Analogous to the tunica adventitia of blood vessels
  - Continuous with the parietal serous layer at the roots of the great vessels to form the pericardial cavity
- **Myocardium (middle layer)**
  - Location of cardiac muscle tissue
  - Makes up most of the heart wall (about 95%)
  - Thinner in atria than in ventricles
  - Analogous to the tunic media of blood vessels
- **Endocardium (innermost layer)**
  - Lined by endothelium
  - Supported by a thin layer of connective tissue (subendothelial connective tissue)





http://humanbiologylab.pbworks.com/w/page/97319205/Heart%20Histology

Endocardium

Subdivided into 3 layers:

1. Inner layer:

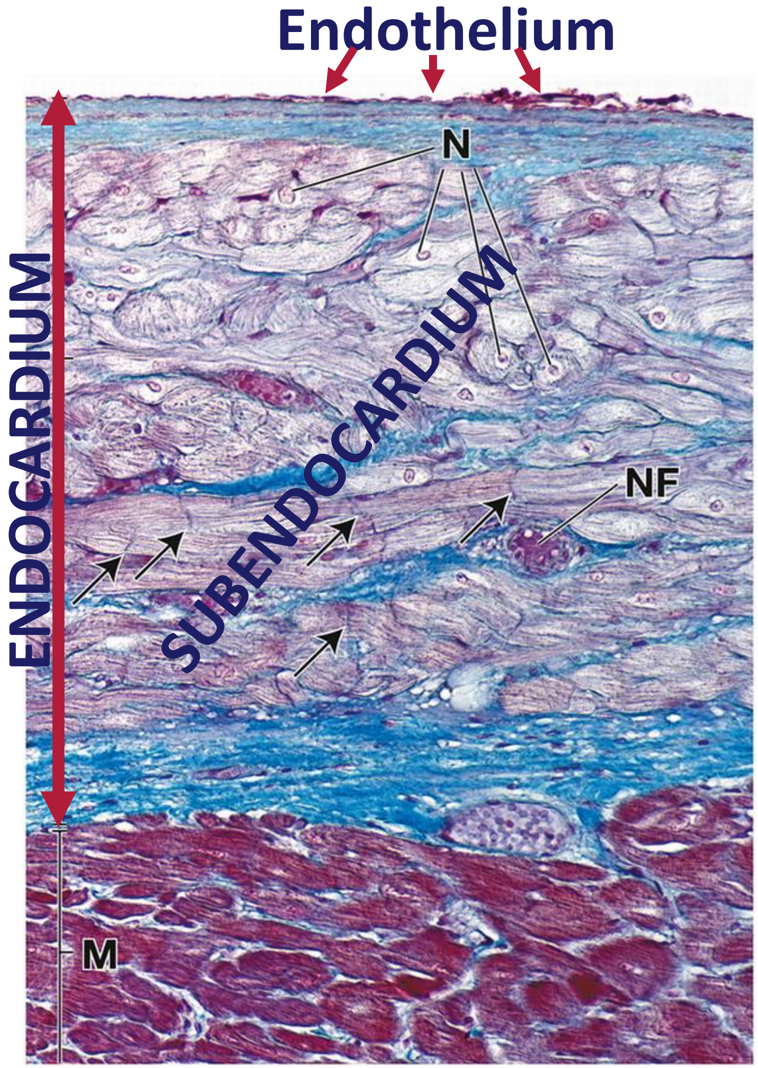
- Composed of simple squamous epithelial cells **(endothelium)**
  - held by tight junctions, and contain gap junctions for diffusion of nutrients
- has an associated underlying **subendothelial** connective tissue (CT)
  - directly beneath the endothelium

2. Middle layer:

- Consists of connective tissue and smooth muscle cells

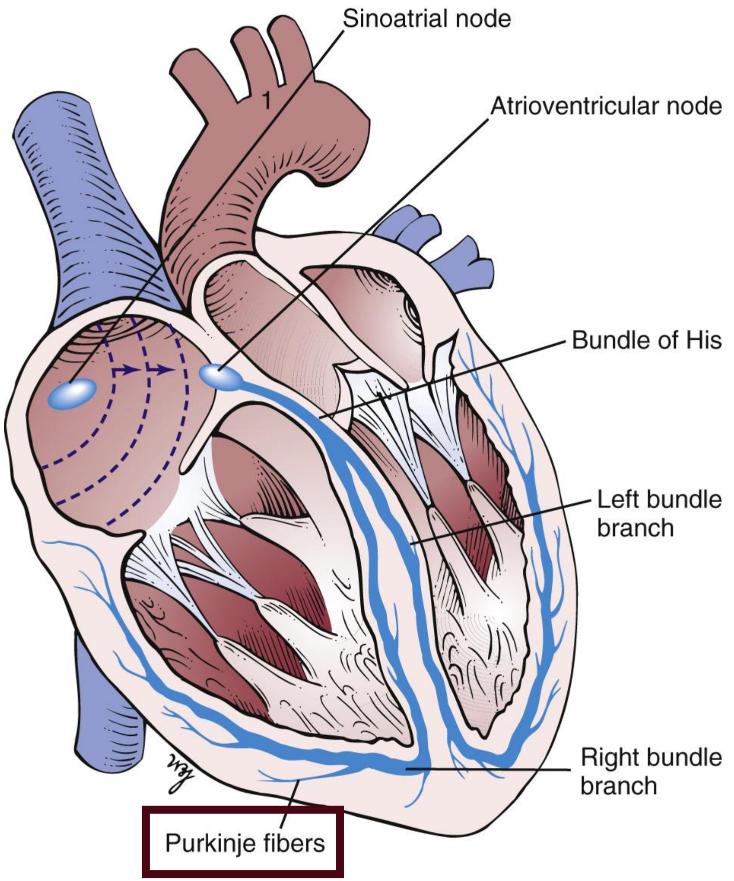
3. Deep layer:

- Composed of connective tissue
- AKA **Subendocardial** layer or subendocardium
  - part of the endocardium deep to the subendothelial connective
  - becomes continuous with the myocardium
  - **site of the conducting system of the heart – Purkinje fibers**



Trichrome Stain

Modified Cardiac Muscle: Purkinje Fibers

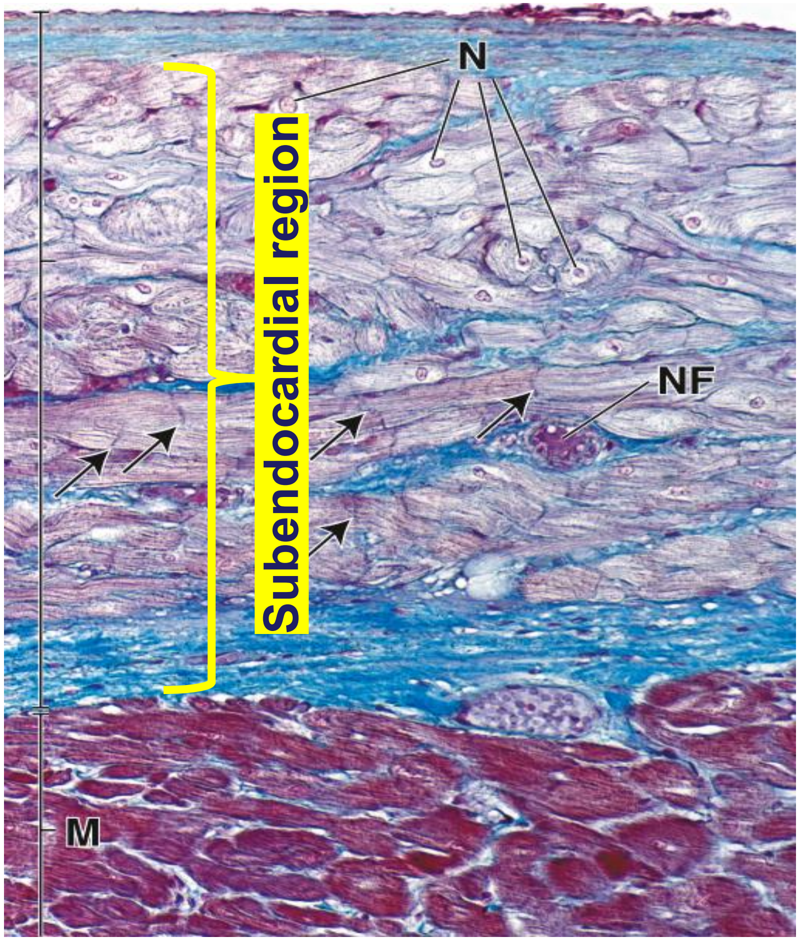


Modified Cardiac Muscle: Purkinje Fibers

- Modified cardiac muscle cells
- Specialized conducting cells
  - Initiate and propagate impulses within the heart
- Larger than ventricular muscle cells
- Located in the ventricular walls within the subendocardial connective tissue

Histological features

- Large, rounded nuclei (much larger than regular cardiac muscle cells)
- Myofibrils located at the periphery of the cell
- Abundant mitochondria
- Pale staining on H&E
  - Large amounts of glycogen (for energy) in the central part of the cell
- Like cardiac muscle cells, they contain intercalated discs which are inconsistent in number and appearance



Mallory-Azan-stain

Arrows: intercalated discs; M - myocardium

NF: autonomic nerve fibers; N: nuclei

Reading Assignment

- Review the Histology Cell Biology portions found in the CPR DLA 1 – Introduction to the Cardiovascular System, regarding the layers of vessel wall and the major components of each layer.

Foundational Concepts – Blood Vessels

- The wall of all arteries and veins contain 3 layers
  - Tunica intima
  - Tunica media
  - Tunica adventitia/externa

  DLA 1
- 4 types of arteries
  - Large arteries eg Aorta
  - Medium arteries eg Brachial
  - Small arteries
  - Arterioles
    - When distinguishing between arteries, we focus on the tunica media
- 4 types of veins
  - Large eg superior vena cava
  - Medium eg brachial vein
  - Small
  - Venules (postcapillary and muscular)
    - When distinguishing between veins, we focus on the tunica adventitia
- 3 types of capillaries
- The wall of blood capillaries contain only the tunica intima
  - When distinguishing between capillaries, we focus on the tunica intima

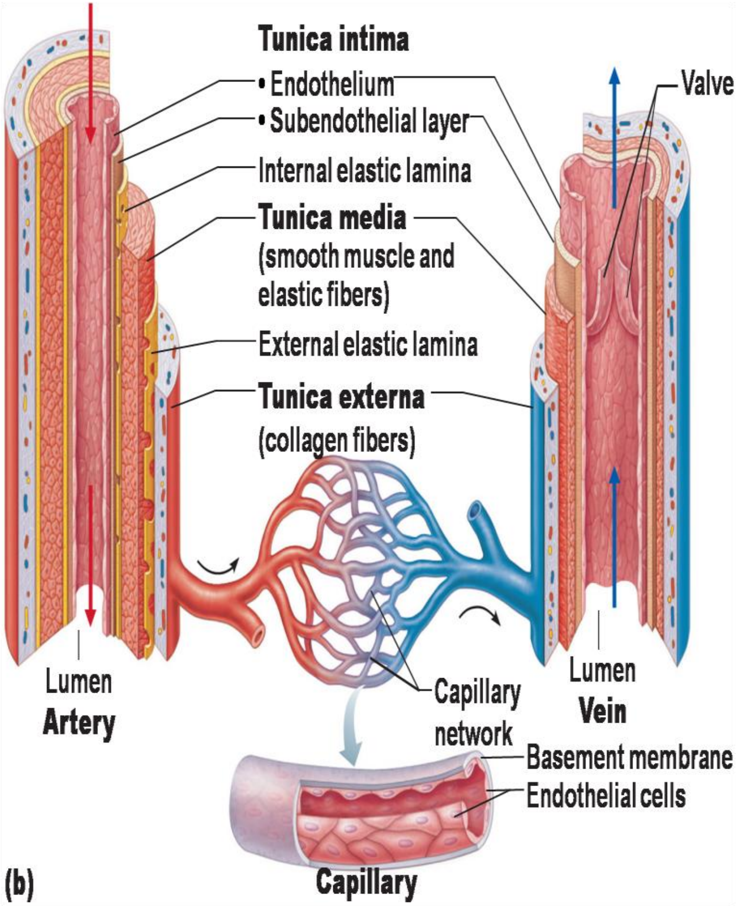
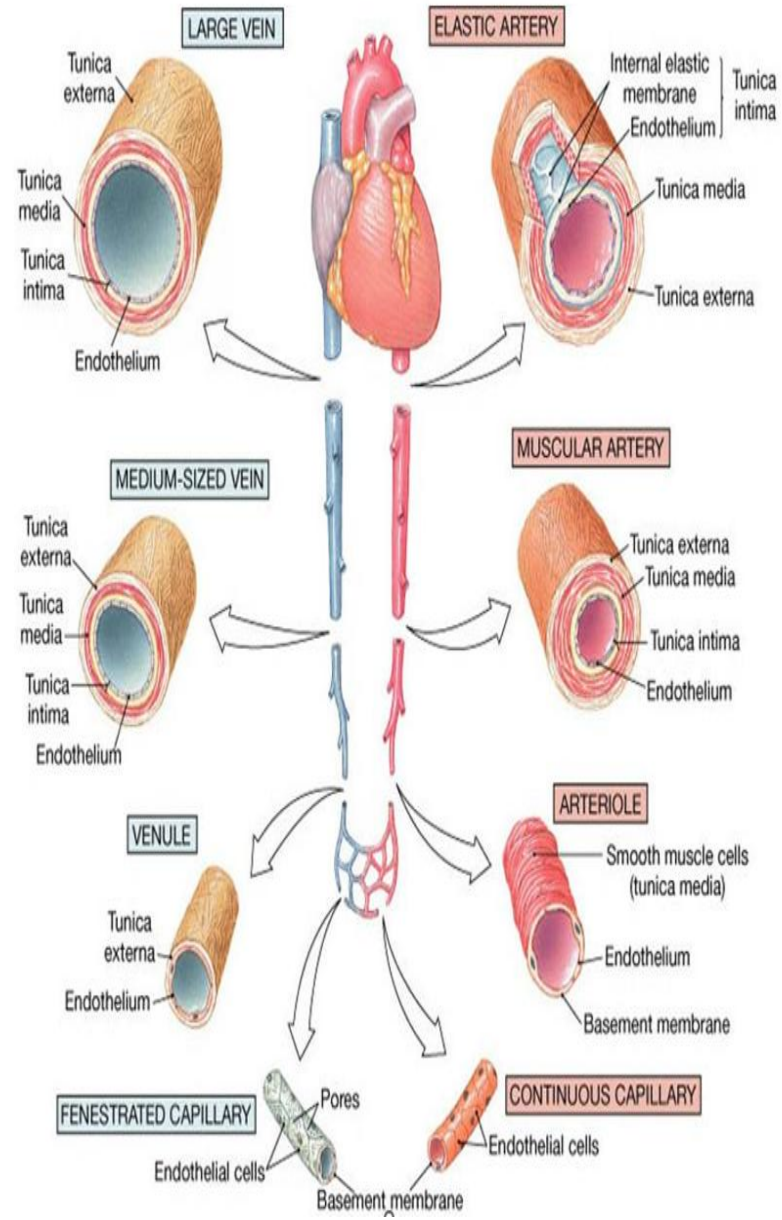


Figure 18.1b

Blood Vessel Histology

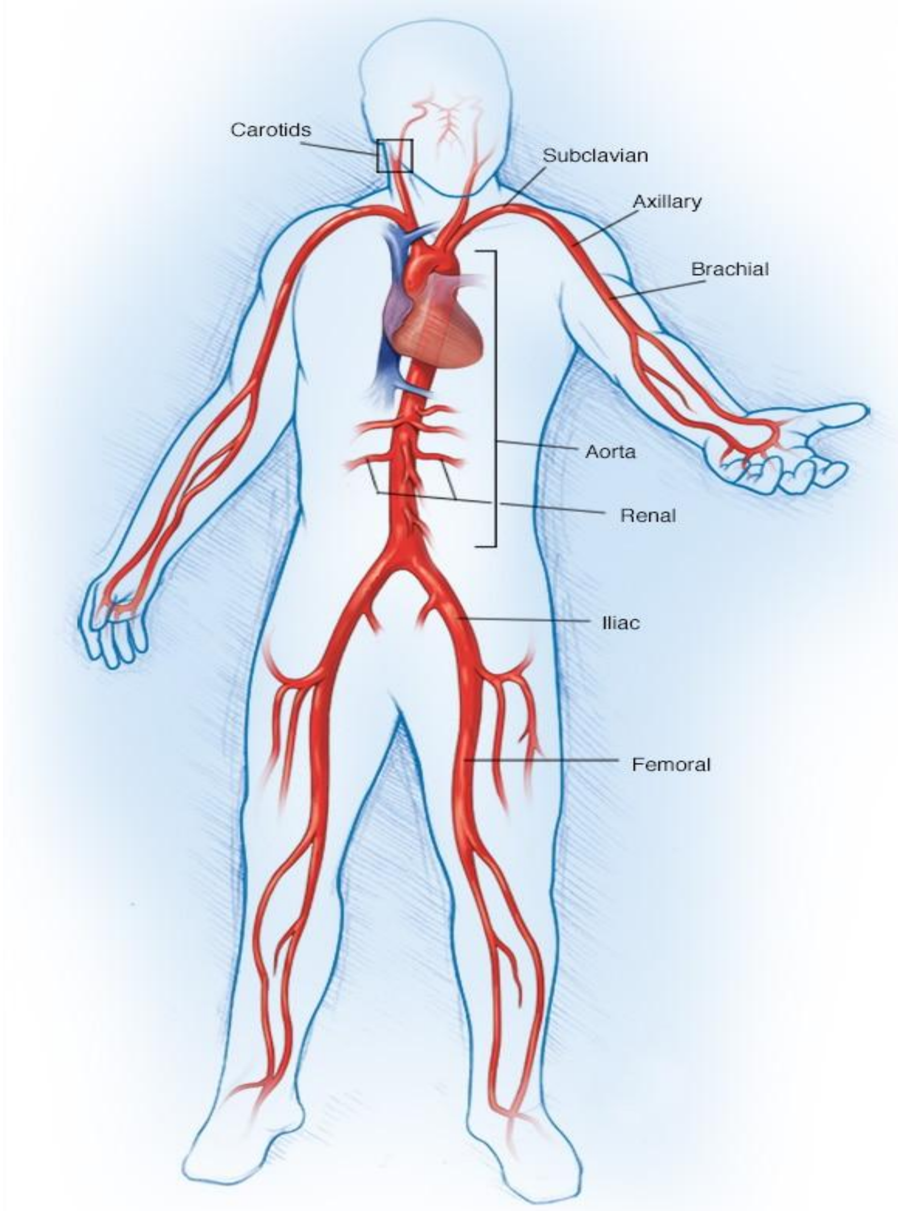
3 major types of vessels:

- Arteries
  - Carries blood away from the heart
    - Mainly oxygenated
  - Examples:
    - arteries (small, medium, large)
    - arterioles
- Capillaries
  - Facilitates exchange
  - Examples:
    - lymphatics (capillaries, vessels)
    - blood capillaries (continuous, discontinuous, sinusoidal)
- Veins
  - Carry blood towards the heart
    - Mainly deoxygenated
  - Examples:
    - veins (small, medium, large)
    - venules (post capillary, muscular)



https://www.earthslab.com/physiology/types-blood-vessels-structure-function-arteries-arte%C2%ACrioles-capillaries-venules-veins/

Blood Vessel Histology: Arteries



The Heart - Source

↓

Elastic (Large) artery
E.g. aorta, subclavian, carotids

↓

Muscular (Medium) artery
E.g. axillary, brachial

↓

Small artery
E.g. anterior interosseus

↓

Arterioles

↓

Capillaries

Large/Elastic Arteries

- Functions to facilitate the continuous and uniformed flow of blood along the tube
  - Expansion of vessel wall to accommodate the large volume of blood during systole
  - Elastic recoiling to maintain intra-vascular pressure

Tunica intima

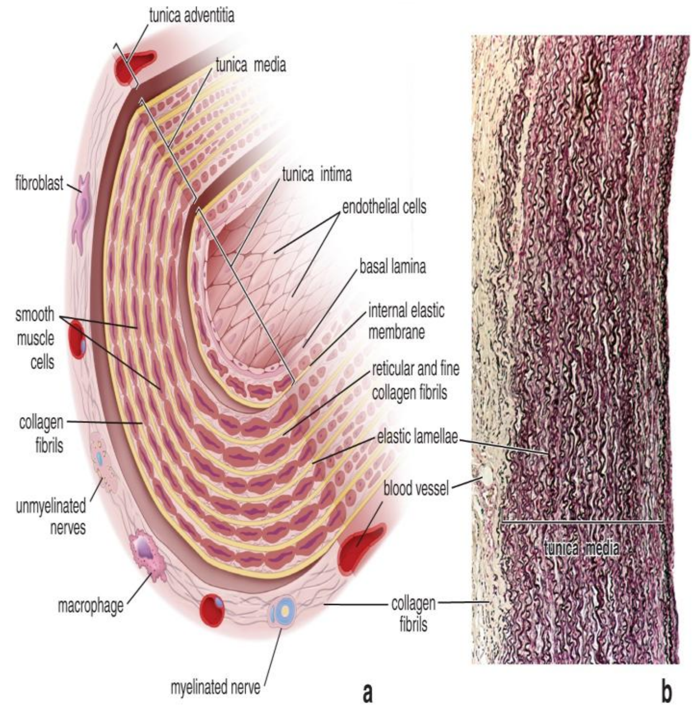
- Contains endothelial cells and its basal lamina
- Underlying subendothelial connective tissue
  - IEL is indistinct from the rest of the vessel wall

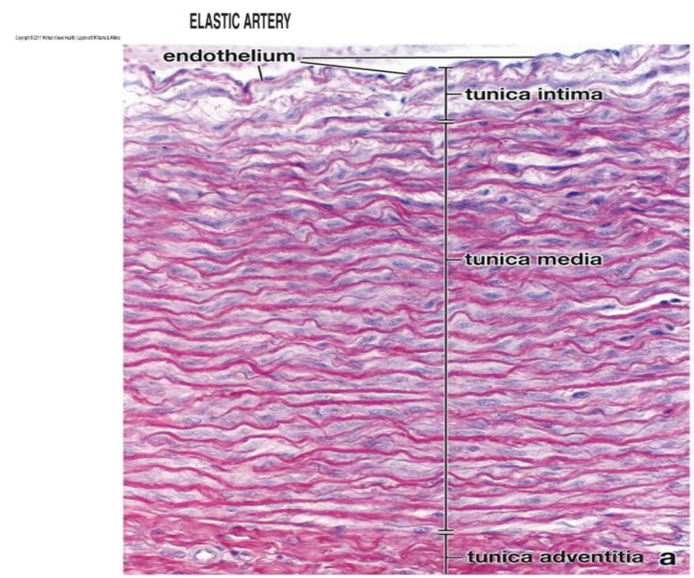
Tunica media

- Thickest layer
- Abundance of **elastic fibers**
  - Arranged in sheets called lamellae between muscle cells
    - **40-70 layers** in an imbricated pattern (like shingles on a roof)
  - Elastic fibers have fenestrations within their lamellae
    - Permit diffusion of nutrients within the vascular wall
- Vascular smooth muscle cells
  - Arranged in layers between the elastic fibers
  - Produce the ECM as no fibroblasts present

Tunica adventitia

- Relatively thin
- See CPR DLA 1 for contents





Medium/Muscular Arteries

- Characteristically has more smooth muscles and less elastic fibers in the tunica media
- Serves as a muscular conduit to distribute blood from elastic vessels to smaller vessels
- Examples: all medium sized vessels that are named – brachial, radial, femoral, tibial etc
- Most abundant arteries in the body

Tunica Intima:

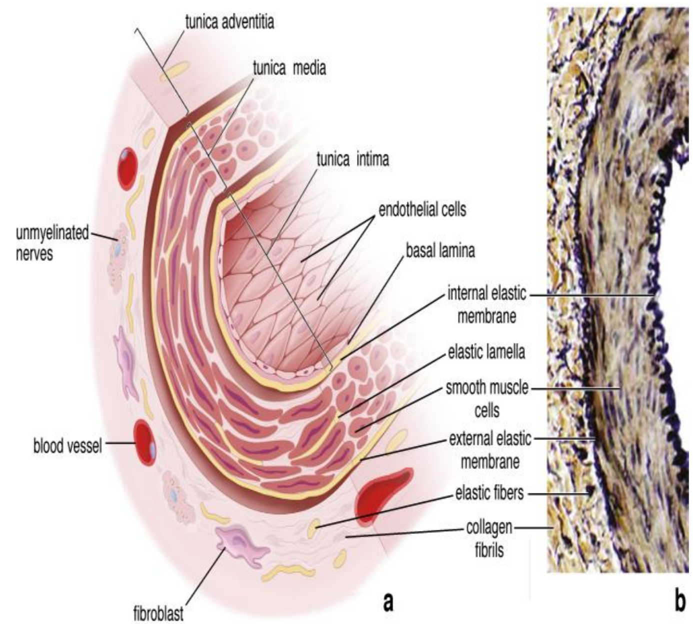
- Thinner - less subendothelial connective tissue
- A **prominent** undulating **internal elastic lamina**/membrane (IEL)

Tunica Media:

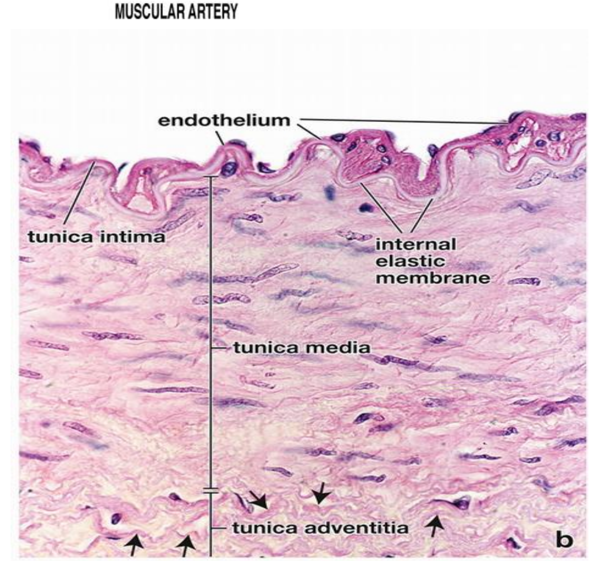
- More **vascular smooth muscles** than elastic fibers
  - Arranged in circular/spiral fashion
    - **8-to-40 layers**
  - Their contraction helps to maintain the blood pressure
- A highly recognizable **external elastic laminae** (EEL) between the tunica media and adventitia

Tunica Adventitia:

- Thicker than elastic arteries
- Collagen is the most predominant fiber



MUSCULAR ARTERY



Small Artery

Tunica intima:

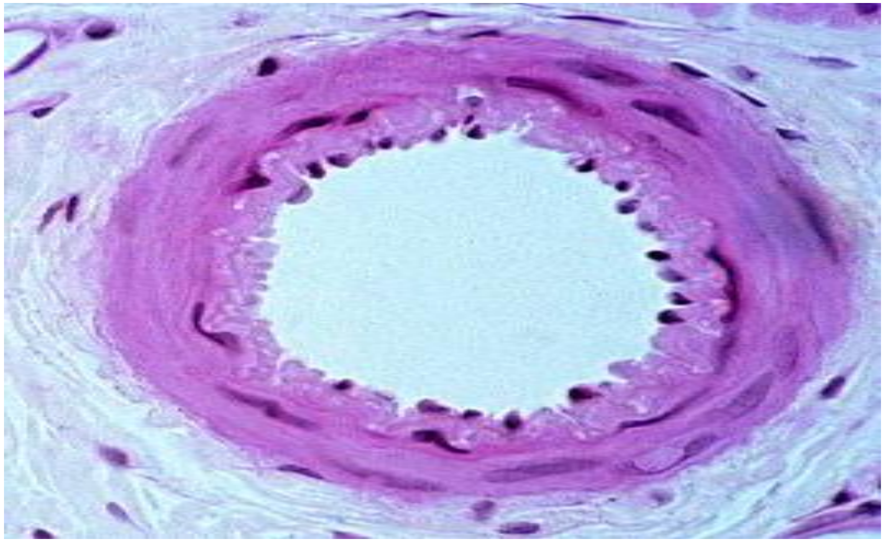
- Minimal subendothelial connective tissue
- IEL present

Tunica Media:

- **3-8 layers** of smooth muscle cells
- Arranged in a circular fashion

Tunica Adventitia:

- Ill-defined connective tissue
- Mainly type I collagen
- Blends in with connective tissue of organs they supply



H&E

Arteriole

- Smallest arteries
  - <0.1 mm in diameter
- Involved in regulating resistance
  - Called resistance vessels
  - Can be closed to generate high resistance to blood flow from arteries into capillaries
    - This is a major determinant of blood pressure

Tunica intima:

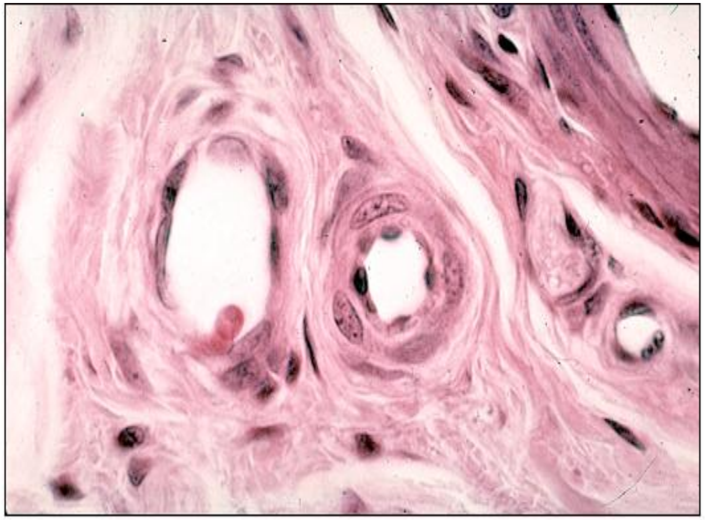
- Very thin
- Endothelium with minimal sub endothelial connective tissue
- IEL may be present or absent

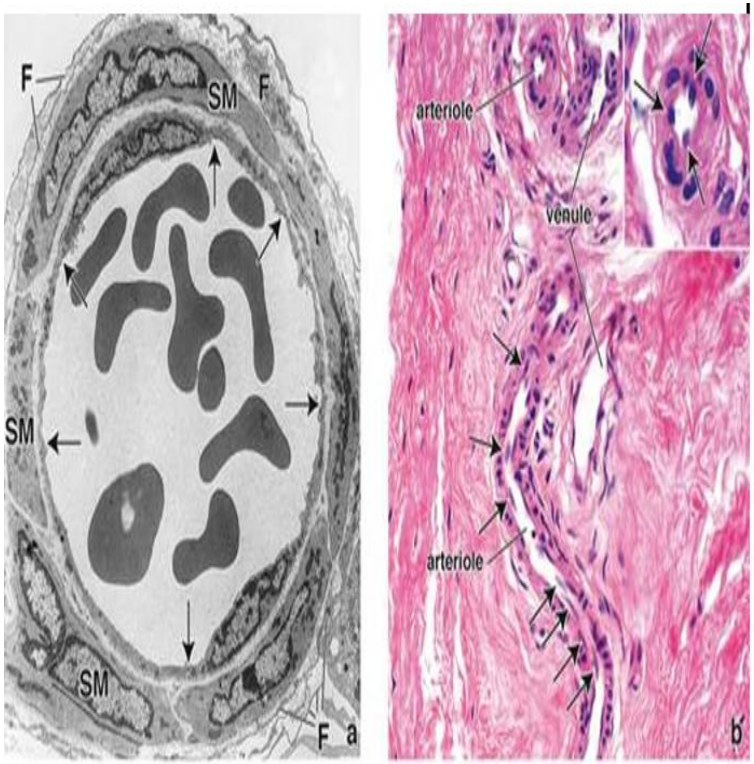
Tunica media:

- 1-2 layers of smooth muscle cells

Tunica adventitia:

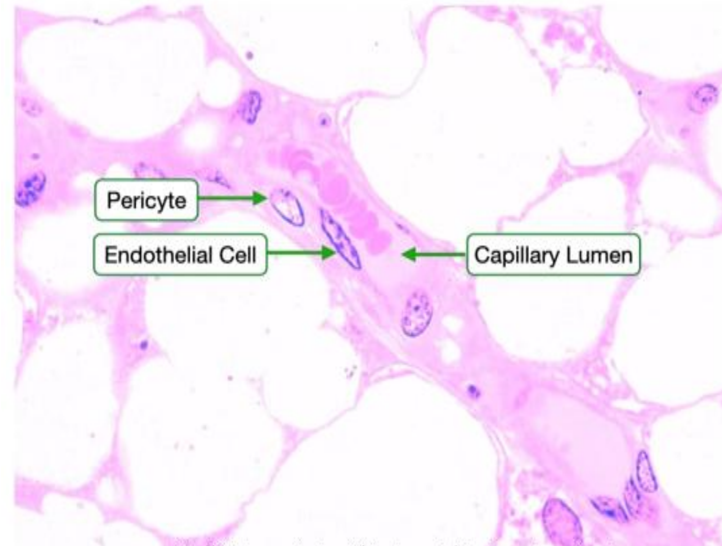
- Ill-defined
- Merges with surrounding connective tissue



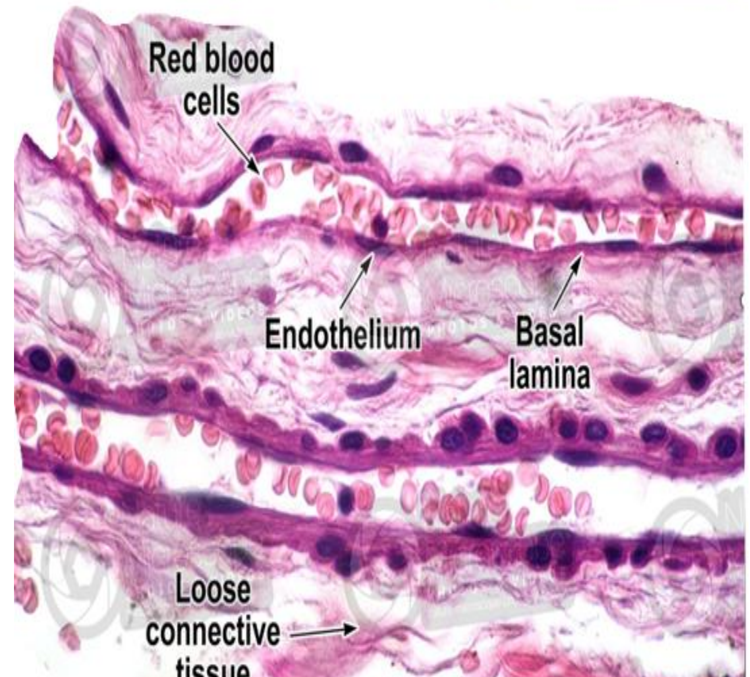


Capillaries

- Smallest diameter of all blood vessels
- Tunica intima only:
  - single layer of endothelial cells and its basal lamina
- Absent tunica media and adventitia
- Extremely thin walls that allows for rapid exchange
- Often associated with perivascular cells eg **Pericytes** (Rouget cells)
  - Acts as stem cells
  - Extend cytoplasmic extensions that encircle the endothelial cells of capillaries
  - Are contractile and responds to NO (nitric oxide) produced by endothelial cells
    - Contract to control capillary blood flow
  - Provides structural support
  - Facilitates bidirectional communication with the CT and the endothelial cells
- Based on their appearance, can be classified into 3 types:
  - Continuous (type I, somatic)
  - Fenestrated (type II, visceral)
  - Discontinuous (type III, sinusoidal)



http://histology.med.yale.edu/blood_vessels/blood_vessels_reading.php



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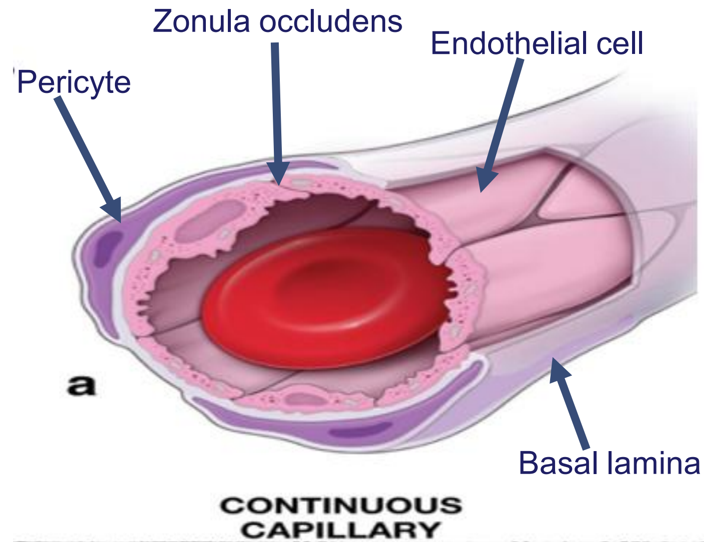
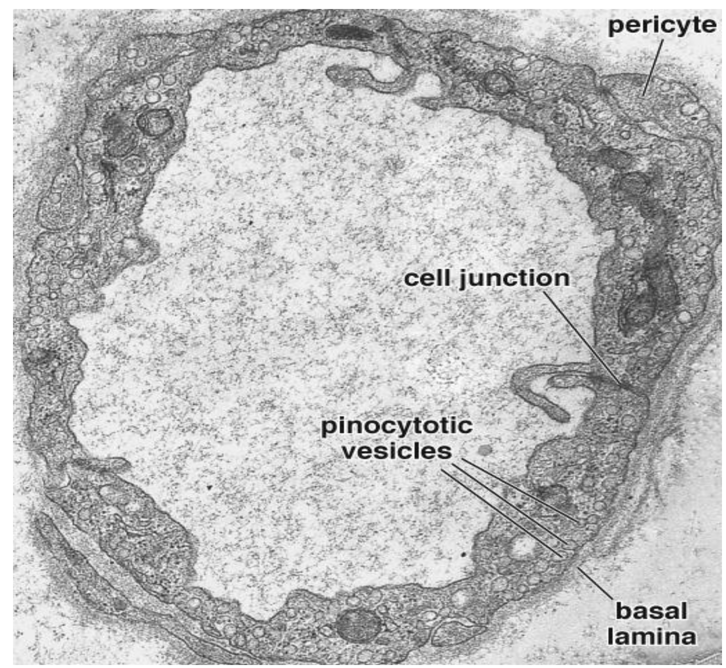
Continuous/Somatic Capillaries

Characterized by:

- Continuous (uninterrupted) endothelium
- Continuous basal lamina
- Endothelial cells are held together by tight junctions
  - Complete control over diffusion, endocytosis and exocytosis
- Endothelial cells contain numerous pinocytic vesicles
- Endothelial cells have few microvilli on luminal surface
- Locations:
  - Connective tissue
  - Nerve tissue
  - Muscle tissue
  - Exocrine glands
  - Vessels that form barriers in organs eg brain, thymus, testis, lung





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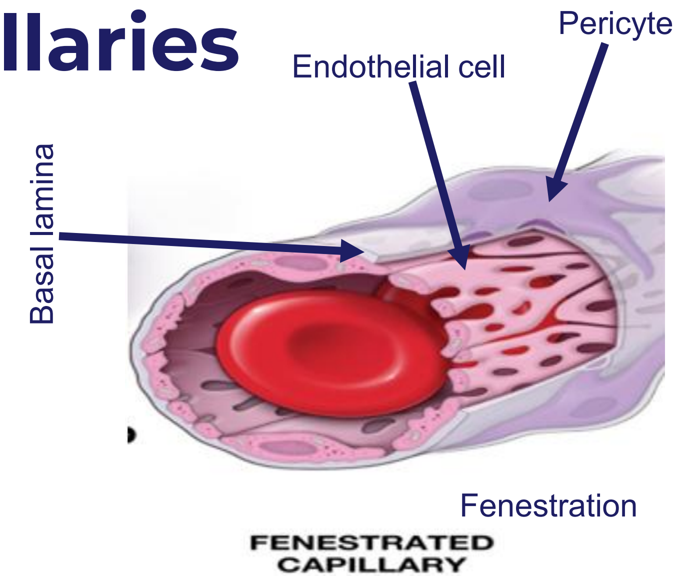
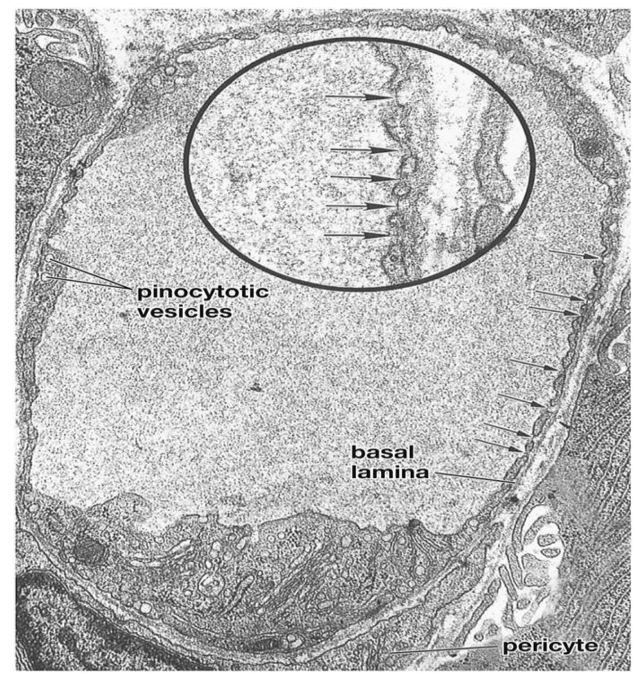
Fenestrated/Visceral Capillaries

Characterized by

- Numerous fenestrations (openings) in the membrane of the endothelial cells
    - Usually 70 – 80 nm in diameter
    - Provides channels across the capillary wall
    - Fenestrations may be covered by a connective tissue diaphragm (except glomerular capillary of kidneys)
- Continuous basal lamina
- Endothelial cells contain numerous pinocytic vesicles
- Endothelial cells also held together by tight junctions
- Locations:
    - Endocrine glands
    - Sites of fluid absorption:
        - Gallbladder
        - Kidney
        - Pancreas
        - Intestinal tract
        - Choroid plexus



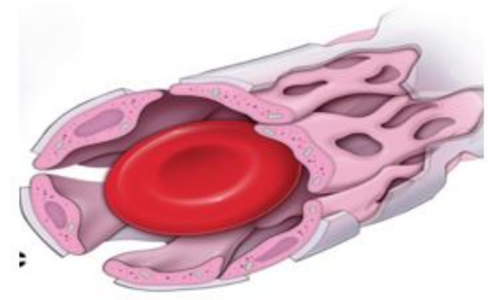


Arrows = fenestrations
Arrows in inset = diaphragm

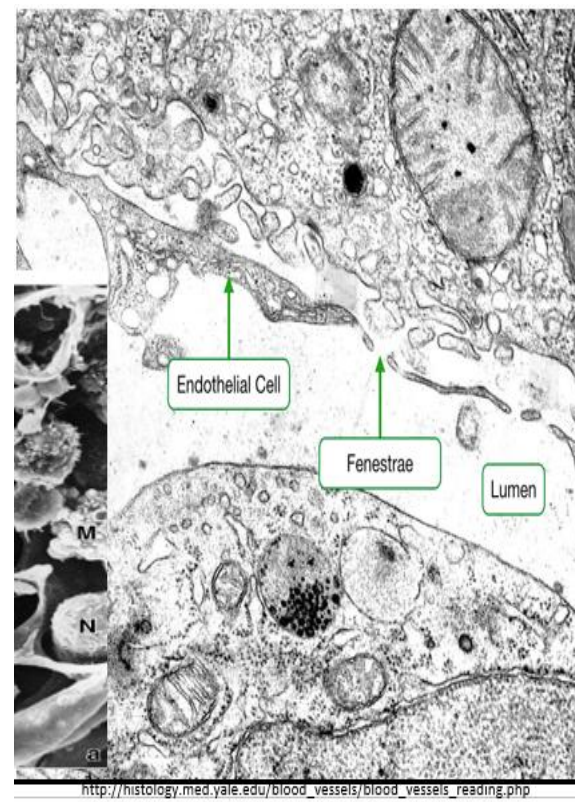
Discontinuous/Sinusoidal Capillaries

Characterized by:

- Large fenestrations in endothelial cell's plasma membrane
- Large intercellular gaps (fenestrae) between endothelial cells
    - Both allow for the passage of blood proteins
    - Absence of tight junctions
    - Fenestrae lack diaphragm
- Discontinuous basal lamina
    - May be absent in some tissues eg spleen
- May not be associated with pericytes in some tissue eg liver, spleen
    - These tissues will have their own specialized endothelial support cell
- Locations:
    - Liver
    - Spleen
    - Bone marrow

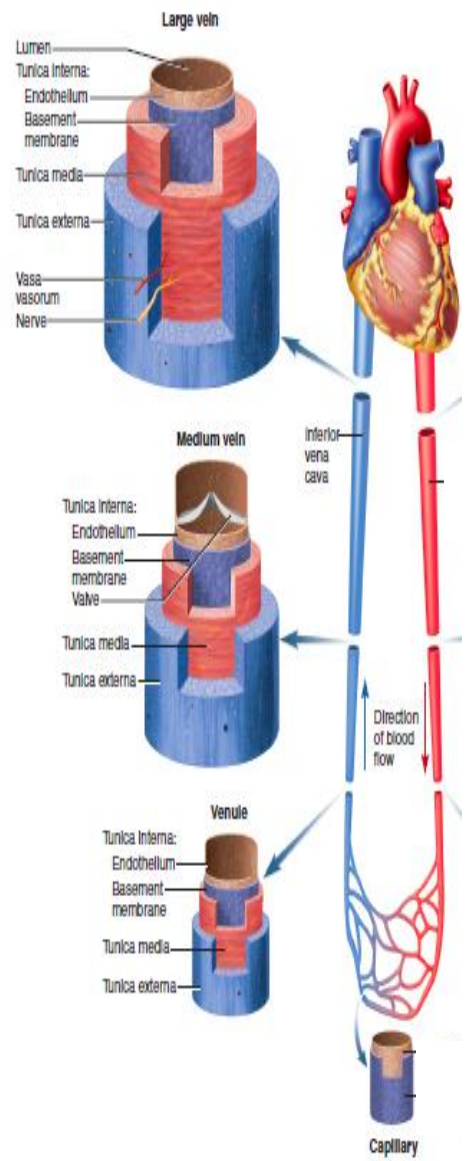


DISCONTINUOUS CAPILLARY



Blood Vessels - Veins

- Carry blood towards the heart
  - Largely deoxygenated blood
- Tunics are present, but not as distinct as arteries
- **4 types** based on size and thickness of tunica adventitia
  - Large
  - Medium
  - Small
  - Venules
    - Further subclassified as postcapillary and muscular venules
    - Luminal diameter of 0.1mm
- Typically thinner walled (less muscular) than arteries with **larger lumen**
- Possess valves
  - Projections of tunica intima



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Heart
↑
Large veins
↑
Medium veins
↑
Small veins
↑
Muscular venules
↑
Post-capillary venules

Venules

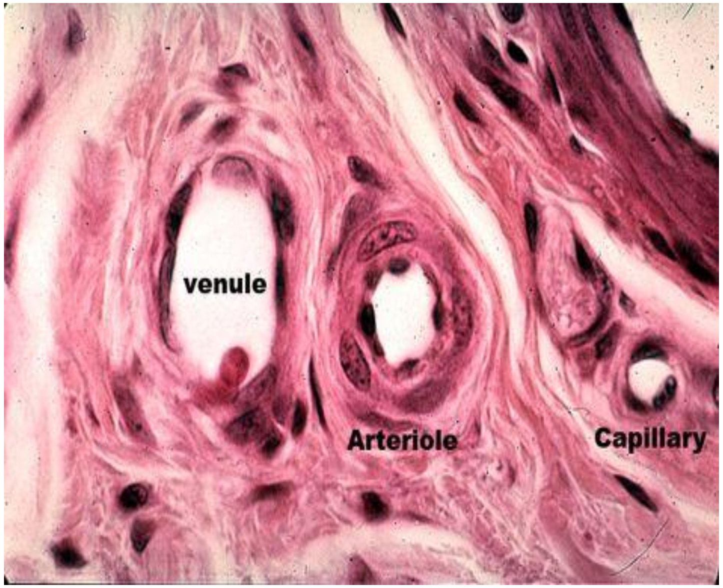
- Two types:
  - Post capillary venules (PCVs) and muscular venules

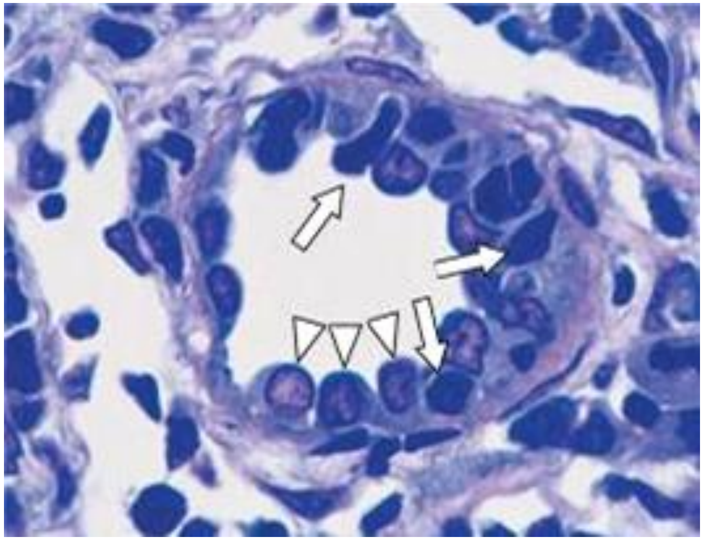
Post capillary venules:

- Collect blood from the capillary bed
- Lacks tunica media and tunica adventitia

Tunica intima:

- Endothelial lining with its basal lamina
- Has pericytes
  - Supportive role
    - More abundant here than in capillaries
  - Communicates with endothelial cells via gap junctions
- Endothelium is the principal site of action of vasoactive agents such as serotonin and histamine
  - Response leads to extravasation of fluids and the migration of immune and inflammatory cells from the circulation into the tissue
- PCVs in lymph nodes are called high endothelial venules (HEV)
  - Lined by cuboidal or columnar endothelial cells
  - Support high levels of lymphocyte migration from the blood





Note the endothelial cells (arrow heads) have high cuboidal shape; these vessels are called HEV (high endothelial vessels)

Venules: Muscular Venules

- Located distal to post capillary venules
- Tend to course with arterioles
- **Distinguished from post capillary venules by the presence of a tunica media**
- Tunica adventitia also present
- Muscular venules also lack pericytes

Tunica media:

- 1-2 layers of vascular smooth muscles

Tunica adventitia:

- Present and very thin

Small veins

- A continuation of muscular venules
- Luminal diameter vary from 0.1mm to 1mm
- Contains all 3 layers/tunics

Tunica intima:

- Features as discussed in the DLA

Tunica media:

- **2-3 layers of vascular smooth muscle**

Tunica adventitia:

- Thicker connective tissue as compared to muscular venules

Medium (Deep) Veins

- Defined as having a diameter of up to 10 mm

Tunica intima

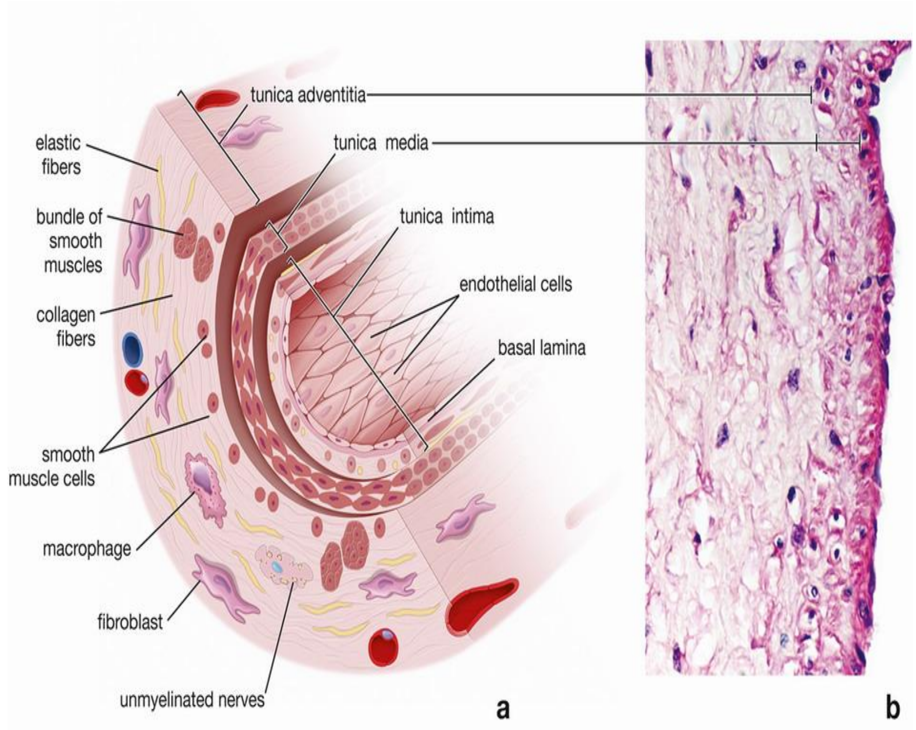
- In some cases a thin, discontinuous IEL
- Other features as outlined in the DLA

Tunica media

- Thinner than its counterpart medium sized arteries

Tunica adventitia:

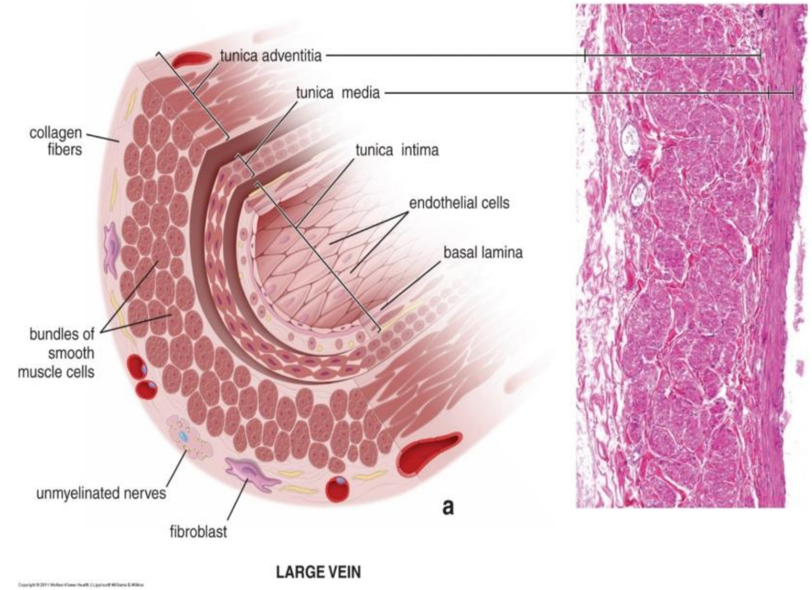
- Thicker than media
- Contains collagen and elastic fibers
- Has bundles of smooth muscles
  - Longitudinally arranged

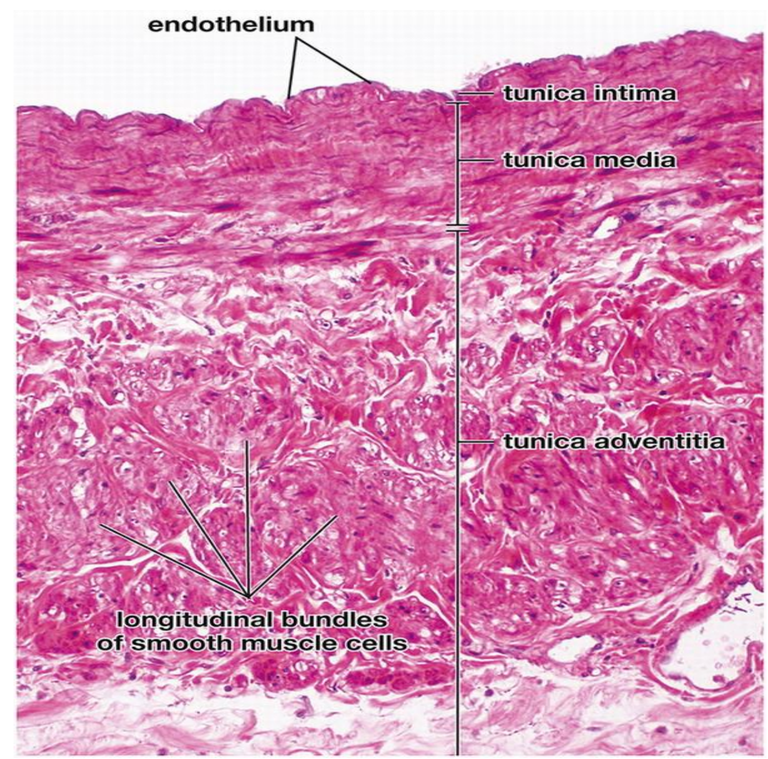


MEDIUM-SIZED VEIN

Large Veins

- Defined by having a diameter 10 mm or greater
- **Tunic intima:**
  - Very thin subendothelial connective tissue
  - See DLA for other features
- **Tunica media:**
  - Relatively thin
  - Contains collagen fibers, fibroblasts and smooth muscle cells
- **Tunica adventitia:**
  - Thickest layer of the vessel wall
  - Longitudinally oriented bundles of smooth muscle cells
- Egs – subclavian, portal, venae cavae

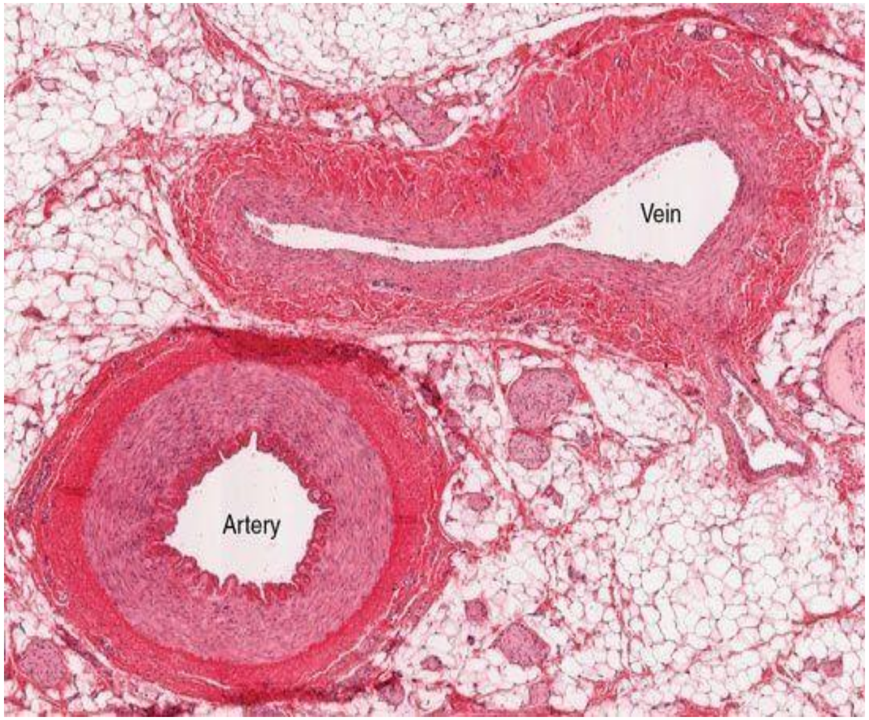




Summary Slide

This slide seeks to highlight major differences between veins and arteries

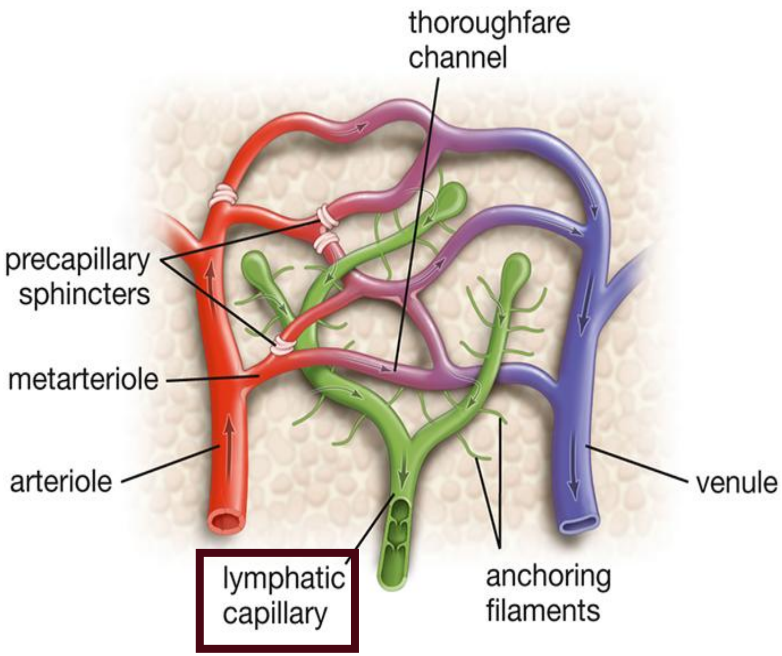
- The **tunica intima** of veins are much **thinner** than that of arteries
- The **tunica media** of veins are much **thinner** than that of arteries and contain very little smooth muscle and elastic fibers
- The **tunica adventitia** of veins is the **thickest layer** and consists of collagen and elastic fibers
- Veins mostly lack the IEL and EEL found in arteries
- Veins have a larger lumen as compared to the accompanying artery

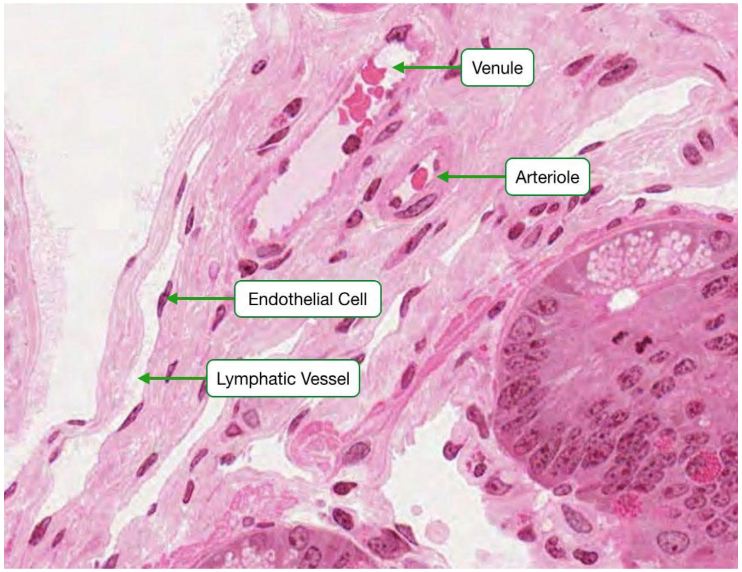


Lymphatic Capillaries vs. Lymphatic Vessels

	Lymphatic Capillary	Lymphatic Vessel
Function	Removal of protein rich fluid from the tissue	Return excess interstitial fluid only from tissues to the bloodstream
Location	Within vascular capillary beds	Connect with chains of lymph nodes eg thoracic duct
Lined by endothelium?	Yes	Yes
Basal lamina	Present but discontinuous	Continuous and surrounded by smooth muscle cells
Zonula Occludens	Present but discontinuous	Present and continuous
Pericytes	Absent	Present
Presence of smooth muscle cells in wall	Absent	Present
Valves	Absent	Present

Lymphatic Vessels





http://histology.med.yale.edu/blood_vessels/blood_vessels_reading.php

Obstruction to lymphatics

Atherosclerosis vs Arteriosclerosis

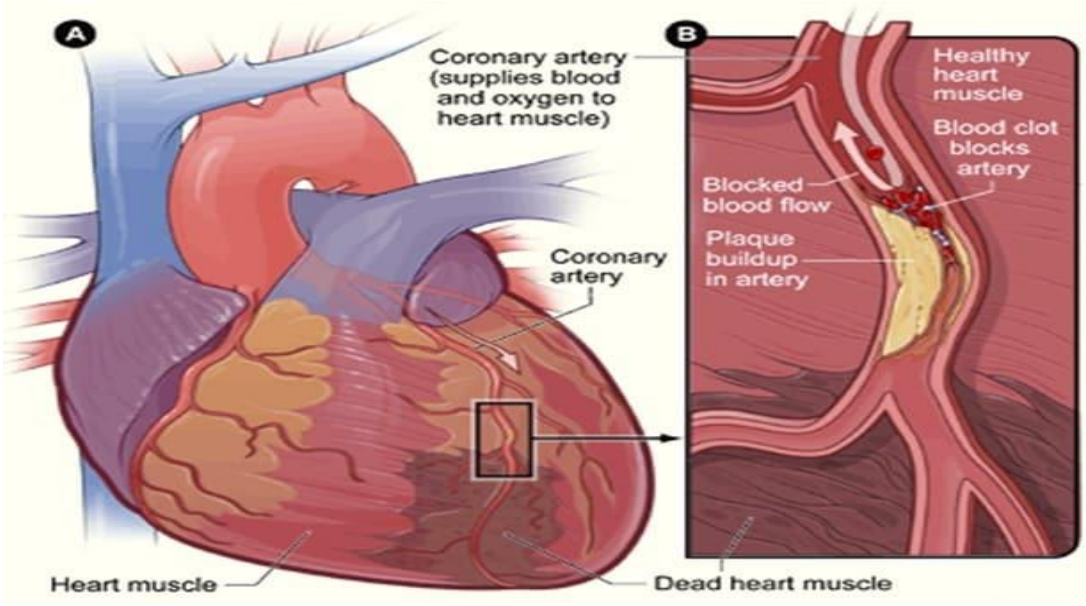
Arteriosclerosis - "Hardening of arterial walls"

- Deposition of substances such as fibrous/scar tissue, proteins, calcium in the arterial walls

- Characterized by loss of elasticity of **T. media**

Atherosclerosis

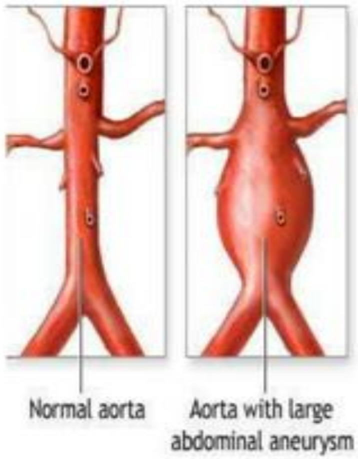
- A type of **arteriosclerosis**
- Characterized by lesions of the **T. Intima**
  - Leads to *atheroma* or *fibro-lipid plaque development which eventually* protrudes into and obstructs the vascular lumen



Aneurysm and Aortic Dissection

Aneurysm

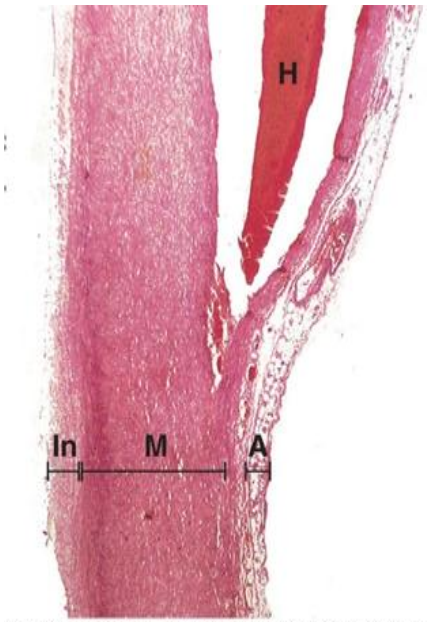
- Ballooning of the vessel wall
- Caused by defective elastic fibers in the tunica media
    - Weakness in the vessel wall with resultant bulging
- Common in the aorta and vessels of the brain
- May be congenital (eg Marfan's Syndrome) or acquired (eg hypertension)



https://ggallk.blogspot.com/2021/0

Aortic Dissection

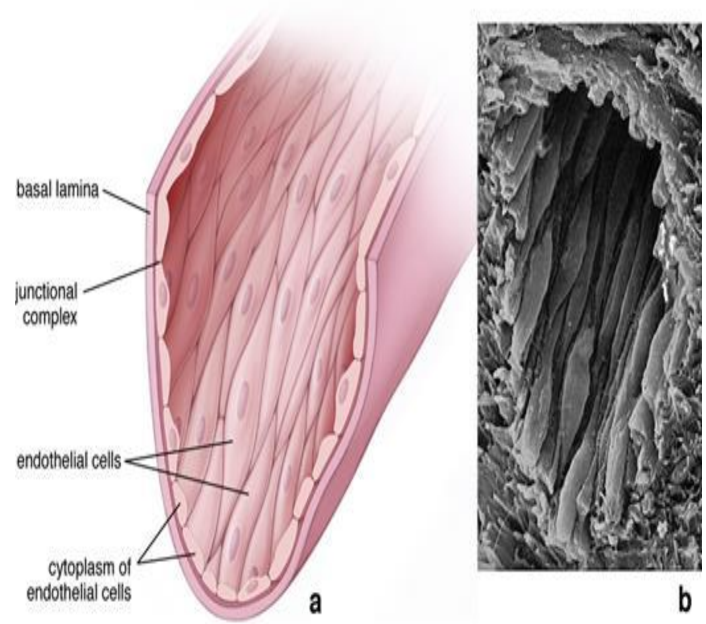
- Blood in tunica media
- Usually due to degeneration of smooth muscle and elastic fibers in the tunica media
- Injury to the tunica intima **In.** leads to tracking of blood into the tunica media **M**
- Separation (dissection) usually happens between the middle and outer thirds of the tunica media
- The hematoma **H,** then frequently bursts through the tunica adventitia **A,** with fatal consequences
- Commonly affects the thoracic aorta



Young et al: Wheater's Basic Pathology: A Text, Atlas and Review of Histopathology, 5th Edition.

Vascular Endothelium: Structure

- Single layer of squamous cells that lines the endocardium and the vessels of the circulatory system
  - Aligned with their long axis parallel to the flow of blood
- Very important in blood homeostasis
- Contribute to the structural and functional integrity of vascular wall
- At the luminal surface, express a variety of surface adhesion molecules and receptors
  - LDL, insulin, histamine
- Functional properties change in response to various stimuli e.g. bacterial antigens, hypoxia, cytotoxins
  - This process is known as **Endothelial Activation**
    - Activation causes endothelial cells to exhibit new surface adhesion molecules and produce different classes of cytokines, lymphokines, growth factors, vasoconstrictors, vasodilators, molecules that control blood coagulation
    - Forms the basis for vascular diseases eg atherosclerosis



Vascular Endothelium: Structure

- Contains Weibel Palade bodies
  - Rod-shaped inclusions within the cytoplasm
    - Store von Willebrand factor
      - A prothrombogenic factor
- In the presence of endothelium damage von Willebrand factor is released
  - Promotes clot formation at the site of injury
  - Prevents excessive blood loss
- Weibel Palade bodies are only present in vascular endothelium of the heart, elastic & muscular arteries, small arteries & arterioles (but NOT present in capillaries)
- Tissue plasminogen-activator inhibitor is another example of a prothrombogenic factor released when there is injury to the endothelial cells

Functions of Endothelial cells

JACC Vol. 60, No. 16, 2012
October 16, 2012:1455–69

Marti *et al.* 1457
Endothelial Function and Heart Failure

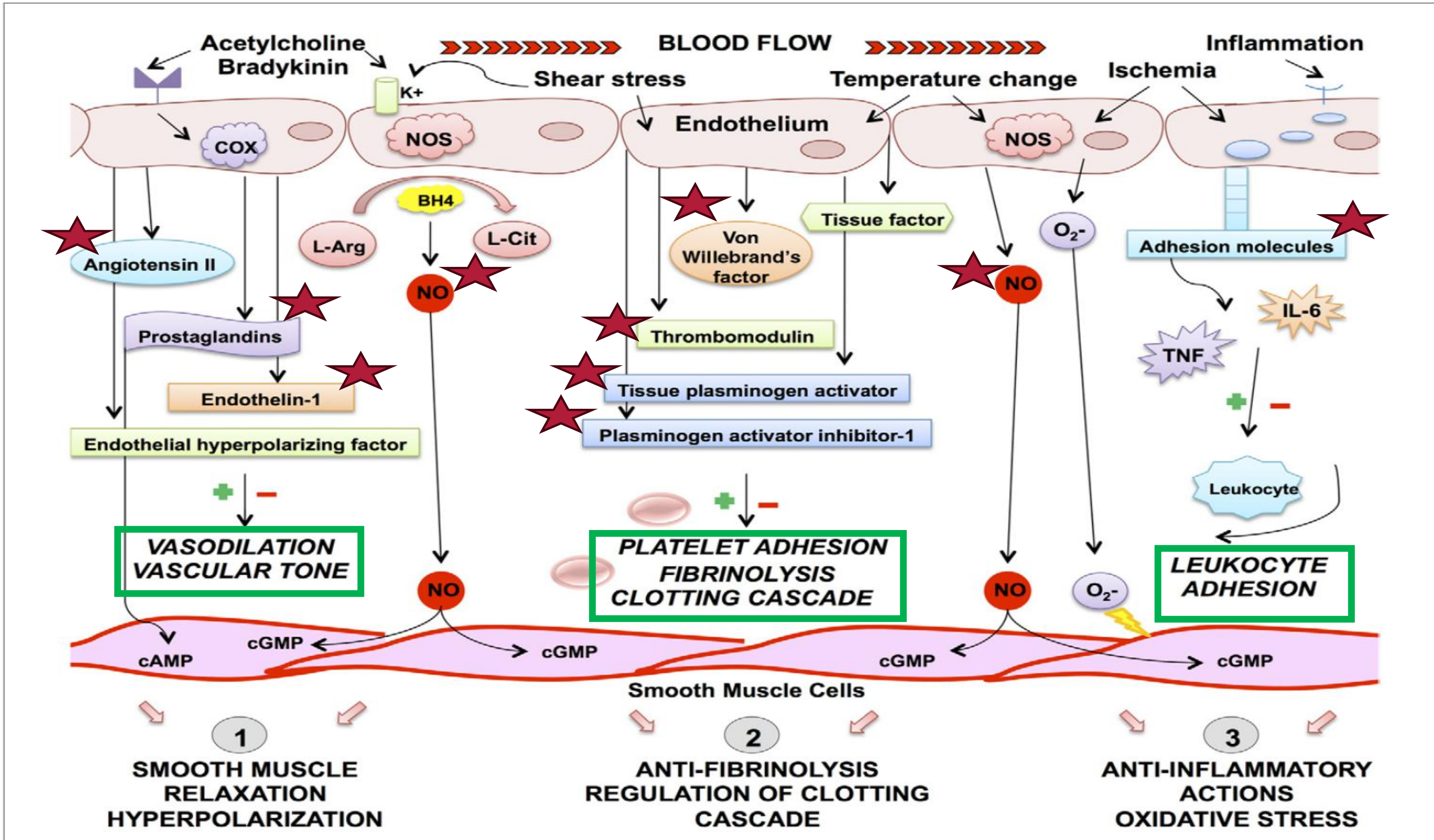


Figure 2 Normal Endothelial Function

Functions of Endothelial cells

- Maintain an **antithrombogenic barrier**
  - Produces and secretes anticoagulants
    - Eg thrombomodulin
      - Inhibit platelet attachment and aggregation
  - Produces and secretes antithrombogenic substances
    - Prevent platelet aggregation and release of factors that cause clot formation
      - Eg prostacyclin, tissue plasminogen activator
  - Surface coated with heparin-like sulfated GAGS
    - Activates antithrombogenic substances
- **Permeability barrier**
  - controls the composition of interstitial tissue fluid
    - allows for simple diffusion and active transport of substances across the endothelial membrane
- Regulation and modulation of **immune response**
  - Interactions of lymphocytes with receptors on endothelial surface
    - Uses adhesion molecules (e.g., selectins, interleukins)
- VEGF (vascular endothelial growth factor)
  - Angiogenesis during embryogenesis, vessel trauma, tumor growth
    - Gives rise to new blood vessels
- Modification of lipoproteins
  - Produce free radicals that oxidize lipoproteins

Functions of Endothelial Cells – Cont'd

- **Modulation of blood flow and systemic vascular resistance**
  - Achieved via the secretion of vasoactive substances
- Vasoactive substances act on the vascular smooth muscle fibers within the tunica media
- Their release result in either vasoconstriction or vasodilation
  - Contraction of the vascular smooth muscle leads to vasoconstriction
  - Relaxation of the vascular smooth muscle leads to vasodilation
    - Vasoconstriction leads to decrease in luminal vessel diameter which in turn leads to increased vascular resistance and ultimately increased systemic arterial pressure
    - The reverse is true for vasodilation
- The table below contains example of vasoactive substances:

Substance Secreted	Effect on blood vessel
Endothelin	Vasoconstriction
Angiotensin Converting Enzyme (ACE)	Vasoconstriction
Prostaglandin H2	Vasoconstriction
Nitric Oxide	Vasodilation
Prostacyclin	Vasodilation